

427 $\Rightarrow$ Adaboost example

Dataset $\Rightarrow$ predicting if the customer will repay a loan

ID	Income	Credit Score	previous default (failure to repay)	Repay	w_i Sample weight initial
1	High	Good	No	Yes	$1/6$
2	High	Poor	No	Yes	$1/6$
3	Low	Good	No	Yes	"
4	Low	Poor	No	No	"
5	Low	Poor	Yes	No	"
6	High	Poor	Yes	No	"

Initially ~~no~~ every sample has same weight/equal weight. $w_i = 1/6$

Step 1

Find the best decision splitter for first Stump. using weighted Gini value

Income:

Income = High
 Sample $\Rightarrow$ 1, 2, 6 (#)
 Target $\Rightarrow$ Y, Y, N
 for yes, total weight =
 $\frac{1}{6} + \frac{1}{6} = \frac{2}{6}$ {2 Yes} = .333
 for No, total weight =
 $\frac{1}{6}$ {1 No} = .166
 Total = $\frac{3}{6} = .5$

weighted
gini
gini

$$= 1 - \left(\frac{.333}{.5} \right)^2 - \left(\frac{.166}{.5} \right)^2 = \approx 0.444$$

*Note, alternatively we just did this before
 $1 - \left(\frac{2}{3} \right)^2 - \left(\frac{1}{3} \right)^2 \Rightarrow$ ans gini. This worked
 because all entries had same weight, the
 answer will same for only this iteration, but
 Adaboost will change the weight soon, so
 not wise to use above formula. $\uparrow$

Poor Income = low

$$\text{weighted Gini} = 0.444$$

now weighted Gini:

$$\begin{aligned} & p(i=\text{high}) \cdot \text{gini}(i=\text{high}) + p(i=\text{low}) \cdot \text{gini}(i=\text{low}) \\ &= \frac{3}{6} \cdot 0.444 + \frac{3}{6} \cdot 0.444 = 0.444 \quad \text{--- (i)} \end{aligned}$$

Similarly for credit score:

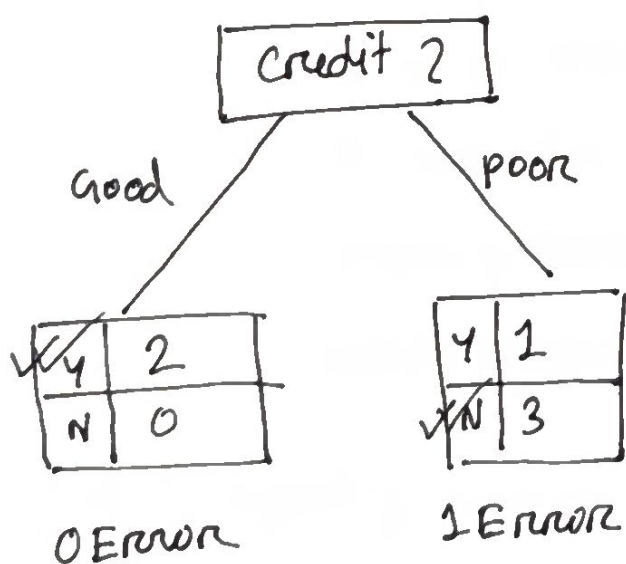
$$\begin{aligned} \text{weighted Gini} &= \frac{2}{6}(0) + \frac{4}{6}(0.375) \\ &= 0.25 \quad \text{--- (ii)} \end{aligned}$$

Similarly, previous default:

$$\text{weighted Gini} = 0.25 \quad \text{--- (iii)}$$

Now comparing (i), (ii), (iii) pick smallest, I am picking credit score.

Decision Stump?



Step 2 Importance value Assign. Not all stump gets the same importance. We have to determine "how much say" a stump will have in the final classification prediction.

Total Error = sum of the weights of incorrect classification.
(ϵ)

$$= 1/6 \quad \left\{ \begin{array}{l} \text{only 1 error, ID 2} \\ \text{weight of that entry} = 1/6 \end{array} \right.$$

now,

Importance score
(α)

$$= \eta \cdot \frac{1}{2} \ln \left(\frac{1-\epsilon}{\epsilon} \right)$$

learning rate

Adaboost is derived by minimizing an exponential loss function that has e^x . So using log base e or known as $\ln$ is more appropriate.

Each stump will have its own importance score to determine how much will it be considered for prediction

$$= (1/2) * \ln \left(\frac{0.8333}{0.1666} \right)$$

$$= 1/2 \ln(5) = 0.8047$$

the range of α is $[-\alpha, \alpha]$

if $\epsilon = .5$; meaning half correct half incorrect basically coin flipping, $\alpha = 0$. meaning no importance at all. For purest subset, $\epsilon = 0$, $\alpha = +\infty$, for all wrong $\epsilon = 1$; $\alpha = -\infty$. ~~Since its hard to calculate infinity in computer code, replacing $\ln$ with $\log$ base 10 can be a good approach.~~

Step 3 updating weights of the entries by the importance score.

④ wrong predicted samples will get max attention for the next stump iteration.

for correct samples $w_i^{\text{new}} = w_i \cdot e^{-\alpha} = (0.166)(.447) = 0.0745$

" wrong " $w_i^{\text{new}} = w_i \cdot e^{\alpha} = (0.166)(2.236) = 0.3727$.

updating the weight in the dataset

ID	...	w_i		normalized w_i
1		0.07		0.10
2		0.37	→ wrong predicted one gets a high weight.	0.50
3		0.07		0.10
4		0.07		0.10
5		0.07		0.10
6		0.07		0.10

⇒ notice, the weight total is not 1, that means it has to be normalized.

$$\text{normalize} = \frac{\text{curr value}}{\text{sum of all value}}$$

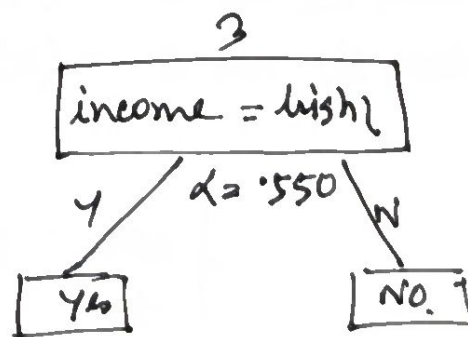
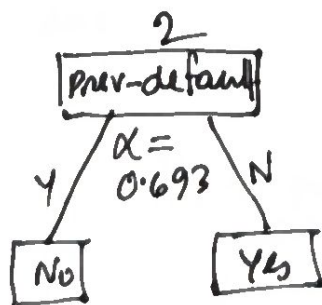
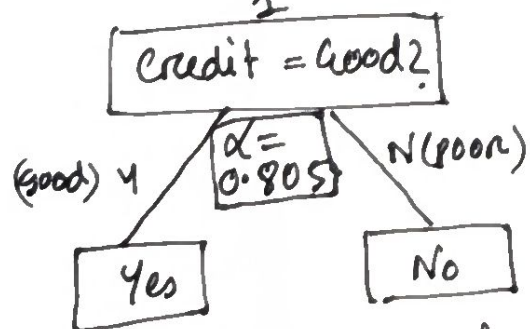
So, for now, repeat step 1 to 3 with this updated weight (ini score now will heavily rely on which decision can make id2 prediction correct as it has high weight).

when to stop: Adaboost stump creation will run until

- fixed number of max weak learners (Stumps)
- perfect classification, $\epsilon = 0$; $\alpha = +\infty$
- weak learner is basically random $\epsilon = 0.5$; $\alpha = 0$
- " " " worse than " $\epsilon > 0.5$; $\alpha < 0$
- validation Accuracy plateau [no significant improvement]

Prediction through Adaboost:

Assume, 3 stumps total:



Predict a ~~for~~ for: { income: high, credit: poor, prev-def: No }

For prediction, sequentially go from one stump to the next. $\text{sum} = 0$

Stump 1 $\rightarrow$ credit: poor $\rightarrow$ prediction: No. For each prediction, if yes:
add ~~1~~ $+\alpha$ and if No: add $-\alpha$.

$\rightarrow$ since, pred: No : $\text{sum} + = (-\alpha)$
 $\Rightarrow \text{sum} = -0.805$.

Stump 2 $\rightarrow$ prev-default: No $\rightarrow$ prediction: Yes.

$\rightarrow$ pred: yes $\Rightarrow \text{sum} + = (+\alpha)$
 $\Rightarrow \text{sum} = -0.805 + 0.693$

Stump 3 $\rightarrow$ income: high $\rightarrow$ prediction: Yes
 $\text{sum} = -0.805 + 0.693 + 0.550$

after all stump, $\text{sum} = 0.438$; if

$\text{sum} > 0$	then final prediction: Yes
$\text{sum} \leq 0$	" " " : NO
$\text{sum} = 0$	means randomly say y/N.

So, prediction Function $F(x) = \sum_{i=1}^{i=n} \alpha_i h_i(x)$; $n = \text{number of stumps}$
 $h_i(x) = +1$ for Yes prediction
 $h_i(x) = -1$ for NO prediction
 $\alpha = \text{importance score of the stump}$

$\left. \begin{array}{l} F(x) > 0 \\ F(x) < 0 \end{array} \right\}$ means final prediction : Yes
" " " : NO